

Elliptic Curves Over Complex Numbers

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Abstract

The goal of this project is to form an understanding of the theory of Elliptic Curves over complex numbers and study an application of the same. We first introduce Elliptic Curves, elliptic (or doubly-periodic) functions and a very useful example of the latter: The Weierstrass ‘ $\wp$ ’ function. The relation between Elliptic Curves, lattices and complex Torus $\mathbf{C}/L$ via the $\wp$ function is studied. Elliptic integrals are introduced and their application in computing periods is explained. Finally an application of elliptic integrals in constructing a conformal mapping from a rectangle to the upper half plane is explained.

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Chapter 1

Pre-Requisites

Complex Number: $z = x + iy$ denotes a complex variable/number.

Real and Imaginary part of z : The real part of z is $\Re z = x$. The imaginary part is $\Im z = y$.

Modulus of z : is defined as $|z| = \sqrt{z\bar{z}}$.

Argument of z : The argument of $z \neq 0$: $\arg z$ is defined as $z = |z|e^{i\arg z}$. It is a multivalued function, as integral multiples of 2π can be added to the angle.

Complex Valued Function/ Mapping: Let $D \subseteq \mathbf{C}$, $h : D \rightarrow \mathbf{C}$ and $h(z) = w$. Here $w = u + iv$ (used in Chapter-6).

All functions like f, g, h , etc. used here are complex valued functions of one variable, unless stated otherwise.

Differentiable Complex Function: $h(z)$ is differentiable at z_0 if

$$\lim_{z \rightarrow z_0} \frac{h(z) - h(z_0)}{z - z_0}$$

exists finitely. This limit is denoted by $h'(z_0)$ and called the derivative of h at z_0 .

Holomorphic/Analytic Function: $h(z)$ is holomorphic (or analytic) in the open set O if its derivative exists and is continuous at each point of O . $h(z)$ is holomorphic at z_0 if it is holomorphic in a “small neighborhood of” (or disc about) z_0 .

Identity Theroem: If h and g are two analytic functions on an open and connected subset of $\mathbf{C}$ and $g = h$ on a subset T of the domain having a limit point, then $f = g$ on

that domain.

Taylor Expansion: A holomorphic function $h(z)$ has a converging power series expansion called the Taylor Expansion:

$$h(z) = \alpha_0 + \alpha_1 (z - z_0) + \alpha_2 (z - z_0)^2 + \dots,$$

in a disk $|z - z_0| < r$.

Taylor Co-Efficients: The Taylor coefficients α_n are given by

$$\alpha_n = \frac{f^{(n)}(z_0)}{n!}$$

$$\implies f(z) = \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!} (z - z_0)^n, \quad |z - z_0| < R(z_0).$$

NOTE: ‘ R ’ is explained below.

Radius Of Convergence Of Power Series: The radius of convergence of the above expansion is given by

$$0 < r \leq R(z_0) := \left[\limsup_{n \rightarrow \infty} |\alpha_n|^{\frac{1}{n}} \right]^{-1}.$$

Zero of Order N : Let $N(\in \mathbf{N})$ be the least number such that $h(z) = 0$ and $h(z) = (z - z_0)^N g(z)$. Here, $g(z)$ is analytic at and $g(z_0) \neq 0$. Such an N is called the order the zero z_0 of h .

Cauchy Integral Theorem¹: Let $h(z)$ be holomorphic in a region D with boundary γ rectifiable².

$$\implies \int_{\gamma} h(z) dz = 0$$

If $z_0 \in D$ then

$$f(z_0) = \frac{1}{2\pi i} \int_{\gamma} \frac{h(z) dz}{z - z_0}.$$

¹Also called Cauchy’s Theorem

²i.e., having finite/measurable length

Entire Function: A function analytic in the whole complex plane $\mathbf{C}$ is said to be entire.

Liouville's Theorem: $h(z)$ be a bounded entire function $\implies h(z) = \text{constant}$.

Isolated Singular Point: Let $z_0 \in O$ (an open set). It is called an isolated singular point of $h(z)$ if $h(z)$ is analytic and also unbounded in a small punctured disk (neighbourhood) of z_0 .

Otherwise, if h is bounded, then z_0 will be called **regular point**.

Laurent Series: Let z_0 be an isolated singular point. Then the Laurent series representing $h(z)$ is:

$$h(z) = \sum_{n=-\infty}^{\infty} \alpha_n (z - z_0)^n,$$

where

$$\alpha_n = \frac{1}{2\pi i} \int_{\gamma_r} \frac{f(z) dz}{(z - z_0)^{n+1}},$$

and γ_r is the circle of radius $r > 0$ with centre z_0 .

Pole: If $\lim_{z \rightarrow z_0} h(z) = \infty$ then z_0 is called a pole of $f(z)$. In this case $\exists m \in \mathbf{Z}$ such that $\alpha_n = 0 \forall n < m < 0$ in the Laurent Series.

OR

$\exists m \in \mathbf{N}$ such that $(z - z_0)^m f(z)$ is analytic in D .

Order Of a Pole: Let m denote order of a pole z_0 .

z_0 is the zero of $1/f(z)$ of order m .

OR The smallest such m in previous definition is called the order of the pole z_0 .

Essential Singularity: z_0 is an essential singularity If the limit $\lim_{z \rightarrow z_0} h(z)$ does not exist.

Meromorphic Function: A function $f(z)$ is said to be meromorphic if it is holomorphic except at the poles. Example: $h(z) = p(z)/q(z)$: both are polynomials in $\mathbf{C}$ and q has zeroes which will be poles of h , $h(z) = \tan z$.

Residue: The constant α_1 in the Laurent series is called the residue of $h(z)$ about

the point z_0 . It is given by

$$\text{Res}_{z=z_0} f(z) = \frac{1}{2\pi i} \int_{\gamma} f(z) dz,$$

where γ is a counterclockwise simple closed contour, not containing any other poles of h . Cauchy's Theorem tells us that the residue of h is zero unless z_0 is a pole.

Residue Theorem:

$$\int_{\gamma} f(z) dz = 2\pi i \sum_{a \in D} \text{Res}_{z=a} f(z)$$

if γ encloses multiple poles a in domain D^3 .

Conformal Mapping: A mapping preserving angles and orientation. Lengths are not necessarily preserved in a conformal mapping. A conformal mapping is one-to-one and onto. See Fig 1.1 for an example.

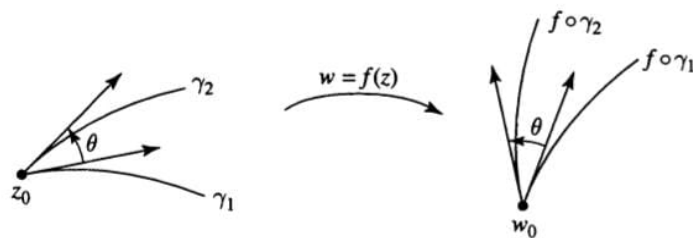


Fig 1.1

Theorem 1.1 If $f(z)$ is analytic at z_0 and $f'(z_0) \neq 0$, then $f(z)$ is conformal at z_0 . Refer [2] for more details.

Winding Number: Let $\gamma(t), a \leq t \leq b$, be a closed path in D . For $z_0 \notin \gamma$, we define the winding number $W(\gamma, z_0)$ of γ around z_0 to be the increase in the argument of $z - z_0$ around γ , divided by 2π . If γ is piecewise smooth, the winding number is the integer (equal to the number of turns made around z_0). It is given by

$$\frac{1}{2\pi i} \int_{\gamma} \frac{dz}{z - z_0}$$

³Domain is a connected open set

Chapter 2

Elliptic Curves

2.1 Definition

A curve of the form $E(K) : y^2 = x^3 + Ax + B$, where A and $B \in$ a field K is called an Elliptic Curve. It is usually denoted as E . This equation is known as the **Weierstrass equation** for an Elliptic Curve.

NOTE: We always add infinity (∞) to an Elliptic Curve for computational purposes, as shown in the next section.

Examples:

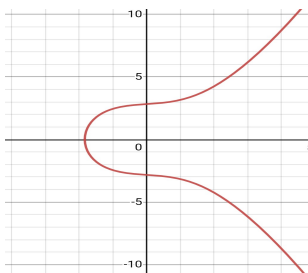


Fig 2.1 $y^2 = x^3 + x + 8$

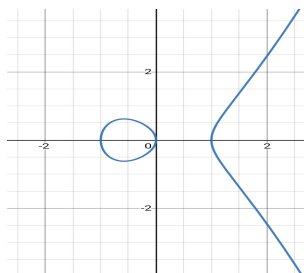


Fig 2.2 $y^2 = x^3 - x$

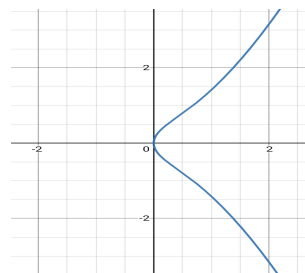


Fig 2.3 $y^2 = x^3 + x$

Here, $y^2 = x^3 - x$ has three distinct real roots and $y^2 = x^3 + x$ has only one real root. If the real roots of the cubic are ρ_1, ρ_2, ρ_3 , then the discriminant of the cubic on the RHS is given as

$$((\rho_1 - \rho_2)(\rho_1 - \rho_3)(\rho_2 - \rho_3))^2 = -(4A^3 + 27B^2).$$

Hence to avoid multiple roots, we shall be assuming everywhere that $4A^3 + 27B^2 \neq 0$.

2.2 Elliptic Curve As A Group

In an Elliptic Curve E , adding two points gives a third point on the same curve [See Fig 2.4 for the case when the points are distinct]. Hence E is an abelian group with identity element as ∞ . Following Theorem gives us details about the same. Refer to Chapter-2 of [3] for a proof and more details.

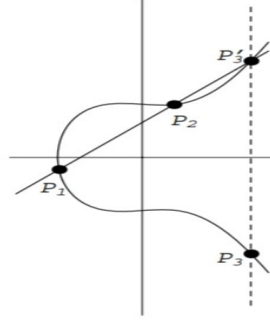


Fig 2.4 Adding Points On An Elliptic Curve

[GROUP LAW] Let $Q_1 = (x_1, y_1)$ and $Q_2 = (x_2, y_2) \in E$ and $Q_1, Q_2 \neq \infty$. Define $Q_1 + Q_2 = Q_3 = (x_3, y_3)$:

1. If $x_1 \neq x_2 \implies x_3 = m^2 - x_1 - x_2, \quad y_3 = m(x_1 - x_3) - y_1, \quad \text{where } m = \frac{y_2 - y_1}{x_2 - x_1}.$
2. If $x_1 = x_2$ but $y_1 \neq y_2 \implies Q_1 + Q_2 = \infty$.
3. If $Q_1 = Q_2$ and $y_1 \neq 0 \implies$

$$x_3 = m^2 - 2x_1, \quad y_3 = m(x_1 - x_3) - y_1, \quad \text{where } m = \frac{3x_1^2 + A}{2y_1}.$$

4. If $Q_1 = Q_2$ and $y_1 = 0 \implies Q_1 + Q_2 = \infty$.

Define

$$Q + \infty = Q$$

for all points Q on E .

2.3 The j -Invariant

j -Invariant. Let $A, B \in K$, a field of characteristic $\neq 2$ or 3 . The j -invariant of E is defined as

$$j = j(E) = 1728 \frac{4A^3}{4A^3 + 27B^2}.$$

NOTE: that the denominator is the negative of the discriminant of the cubic polynomial in x . Therefore it is non-zero by our assumption of section 2.1. Characteristic is chosen not to be 2 or 3 to avoid the zero j -invariant.

The j -invariant tells us relation between two Elliptic Curves. This is shown in the next Theorem.

Theorem 2.2 Consider two Elliptic Curves $y_i^2 = x_i^3 + A_i x_i + B_i, i = 1, 2$. with j -invariants j_i . and Then $j_1 = j_2 \implies \exists \lambda \neq 0$ in $\bar{K}^1$ such that

$$A_2 = \lambda^4 A_1, \quad B_2 = \lambda^6 B_1.$$

The following transformation changes one equation to the other

$$x_2 = \lambda^2 x_1, \quad y_2 = \lambda^3 y_1. \tag{2.1}$$

Proof. **Case-(i)** $A_1 \neq 0$:

$\implies j_1 \neq 0$. Hence $A_2 \neq 0$ as $j_1 = j_2$. Choose λ such that $A_2 = \lambda^4 A_1$.

$$\begin{aligned} \implies \frac{4A_2^3}{4A_2^3 + 27B_2^2} &= \frac{4A_1^3}{4A_1^3 + 27B_1^2} = \frac{4\lambda^{-12}A_2^3}{4\lambda^{-12}A_2^3 + 27B_1^2} = \frac{4A_2^3}{4A_2^3 + 27\lambda^{12}B_1^2}, \\ &\implies B_2^2 = (\lambda^6 B_1)^2 \end{aligned}$$

Therefore $B_2 = \pm \lambda^6 B_1$. If $B_2 = \lambda^6 B_1$, we're done.

If $B_2 = -\lambda^6 B_1$, λ is made $i\lambda$.

Hence the relation $A_2 = \lambda^4 A_1$ is preserved and also $B_2 = \lambda^6 B_1$.

Case (ii) $A_1 = 0$

$\implies A_2 = 0$. $4A_i^3 + 27B_i^2 \neq 0$ for $i = 1, 2$. Therefore $B_1, B_2 \neq 0$. We Choose λ giving $B_2 = \lambda^6 B_1$.

□

¹the algebraic closure of K

Remark. However, the j -invariant can tell us about the isomorphism relation between two Elliptic Curves only when the field is algebraically closed. If K is a non-algebraically closed field, two curves with the same j -invariant might not be transformable to one another other by using only the rational functions with co-efficients $\in K$. Refer Chapter-2 of [3] for more details.

2.4 Elliptic Curves Over $\mathbf{C}$.

Lattice. Consider two linearly independent complex numbers over $\mathbf{R} : \mathbf{w}_1, \mathbf{w}_2$. Then $L = Z\mathbf{w}_1 + Z\mathbf{w}_2 = \{n_1\mathbf{w}_1 + n_2\mathbf{w}_2 | n_1, n_2 \in \mathbf{Z}\}$ is called a lattice.

Fundamental Parallelogram. The set $F = \{a_1\mathbf{w}_1 + a_2\mathbf{w}_2 | 0 \leq a_i < 1, i = 1, 2\}$ is called a fundamental parallelogram for L .

A function on $\mathbf{C}/L$ can be considered on the domain $\mathbf{C}$ satisfying $f(z + \mathbf{w}) = f(z) \forall z \in \mathbf{C}$ and all $\mathbf{w} \in L$.

Torus. We obtain a complex Torus by $\mathbf{C}/L$, where L is a lattice. An Elliptic Curve over $\mathbf{C}$ is isomorphic to a complex Torus. We now study elliptic functions to aid us in proving the same.

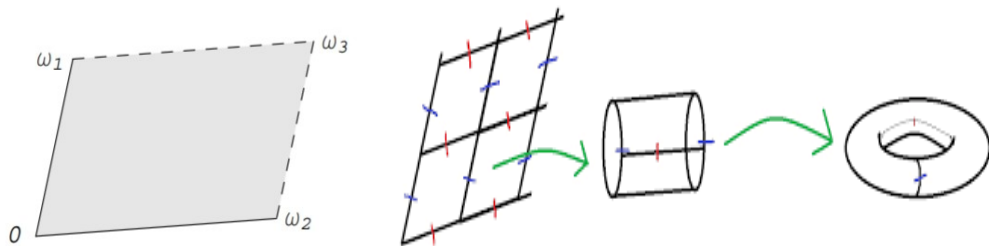


Fig 2.5 The Fundamental Parallelogram and Obtaining Torus From a Lattice

We shall be subsequently denoting $\mathbf{w}_3 = \mathbf{w}_1 + \mathbf{w}_2$.

Chapter 3

Elliptic Functions

Elliptic Functions are doubly periodic meromorphic functions. They originally arose in calculating inverses of elliptic integrals and have some very useful properties, some of which we shall be studying. The first reported study of such integrals was in 1655 when John Wallis began to study the arc length of an ellipse. In 1825, Abel gave an important paper revealing the double periodicity of the elliptic functions. More details regarding historical perspective can be found in [5].

3.1 Existence Of Periodic Functions

Periodic Function. Let $h(z)$ be a meromorphic function. If $\exists w$ (a constant) such that at each z , which is a regular point of h $h(z + \mathbf{w}) = h(z)$. Then $\mathbf{w}$ is called a **period** of h . If $\exists$ such non-zero (non-trivial) $\mathbf{w}$ then $h(z)$ is called periodic. $\mathcal{T}(h)$ denotes the set of all periods of h .

Example. $f(z) = e^{2\pi iz/\mathbf{w}}$. Clearly,

$$\mathcal{T}(e^{2\pi iz/\mathbf{w}}) = \mathbf{w}\mathbf{Z}.$$

Hence $\mathbf{w}$ generates $\mathcal{T}(e^{2\pi iz/\mathbf{w}})$ as every other period is an integral multiple of $\mathbf{w}$. Such a function is called a **simply periodic** function and $\mathbf{w}$ is called its **primitive period**.

There also exist functions with $n > 1$ primitive periods, as shown in Theorem 3.1. Proof can be found in [1].

The condition for $n > 1$ primitive periods is given as:

1. Every other period is an integral linear combination of these n periods, and
2. All n periods are linearly independent over $\mathbf{R}$.

Theorem 3.1 [JACOBI Theorem] There does not exist a non-constant function g with $n \geq 3$ primitive periods. If g is non-constant and $\mathbf{w}, \mathbf{w}'$ are two primitive periods of g , then

$$\Im \frac{\mathbf{w}'}{\mathbf{w}} \neq 0.$$

Example.

$$\phi(z) = \sum_{m=-\infty}^{\infty} e^{(m^2\beta+2mz)\pi i} \quad (3.1)$$

. Let $q = e^{i\pi\beta}$, $\beta \in \mathbf{C}$. The Fourier Expansion of (3.1) gives

$$\phi(z) = 1 + 2q \cos 2\pi z + 2q^4 \cos 4\pi z + 2q^9 \cos 6\pi z + \dots$$

Hence ϕ is an entire function of z having a period as 1.

Also note:

$$\begin{aligned} \phi(z + \beta) &= \sum_{m=-\infty}^{\infty} e^{(m^2\beta+2mz+2m\beta)\pi i} \\ &= e^{-\pi i(\beta+2z)} \sum_{m=-\infty}^{\infty} e^{[(m+1)^2\beta+2(m+1)z]\pi i} \\ &= e^{-\pi i(\beta+2z)} \sum_{n=-\infty}^{\infty} e^{[n^2\beta+2nz]\pi i}. \end{aligned}$$

$$\implies \phi(z + \beta) = e^{-\pi i(\beta+2z)} \phi(z). \quad (3.2)$$

Taking Logarithm of [3.2] followed by double derivative w.r.t. z of both sides :

$$\frac{d^2}{dv^2} \ln \phi(z + \beta) = \frac{d^2}{dv^2} \ln \phi(z)$$

and (as ϕ has period 1 also):

$$\frac{d^2}{dv^2} \ln \phi(z + 1) = \frac{d^2}{dv^2} \ln \phi(z).$$

$\implies \Psi(z) := \frac{d^2}{dv^2} \ln \phi(z)$ is an example of a doubly periodic function with periods β and 1 and $1/\beta \notin \mathbf{R}$.

Also Ψ is meromorphic, as $\phi(z)$ is an entire function. Hence the only singularities of Ψ are simple poles. These are zeroes of the function ϕ . Therefore the only singularities we find are poles of order 2.

Such doubly periodic meromorphic functions are called **elliptic functions**. We explain them further below.

3.2 Elliptic Functions and Properties

Elliptic Function. A function $f : \mathbf{C} \rightarrow \mathbf{C} \cup \{\infty\}$ such that $f(z + \mathbf{w}) = f(z) \forall z \in \mathbf{C}$ and all $\mathbf{w} \in L$ is called an elliptic (or doubly periodic function).

We usually just consider $f(z + \mathbf{w}_i) = f(z)$, $i = 1, 2$ as $\mathbf{w}_i$ are the primitive periods.

The Laurent Series expansion of such a non-zero function is:

$$f(z) = \alpha_r(z - w)^r + \alpha_{r+1}(z - w)^{r+1} + \dots$$

with $\alpha_r \neq 0$, $r \in \mathbf{Z}$. Elements of L are called the **periods** of f .

Define for f the order at $\mathbf{w}$ the following as:

Order:

$$r = \text{ord}_{\mathbf{w}} f$$

.

Residue:

$$\alpha_{-1} = \text{Res}_{\mathbf{w}} f.$$

As f is doubly periodic here, $\implies \text{ord}_{w+\mathbf{w}} f = \text{ord}_w f$ and $\text{Res}_{w+\mathbf{w}} f = \text{Res}_w f \forall \mathbf{w} \in L$.

Divisor. The sum of points: $D = n_1[w_1] + n_2[w_2] + \dots + n_k[w_k]$ where $n_i \in \mathbf{Z}$ and $w_i \in F$ is a divisor D . $\deg(D) = \sum n_i$ is the degree of a divisor.

Divisor Of a Function. the divisor of a function f is

$$\operatorname{div}(f) = \sum_{w \in F} (\operatorname{ord}_w f) [w].$$

We now have the definitions to prove the following properties of elliptic functions.

Theorem 3.2 Let f be a doubly periodic function with lattice L and F be the fundamental parallelogram for L . Then the following hold:

1. If f has no poles, $\implies f$ is constant.
2. $\sum_{\mathbf{w} \in F} \operatorname{Res}_{\mathbf{w}} f = 0$.
3. If f is not identically 0 then $\deg(\operatorname{div}(f)) = \sum_{\mathbf{w} \in F} \operatorname{ord}_{\mathbf{w}} f = 0$.
4. If f is not identically 0, $\sum_{\mathbf{w}} \operatorname{ord}_{\mathbf{w}} f \in L$.
5. If f is not constant, then $f : \mathbf{C} \rightarrow \mathbf{C} \cup \{\infty\}$ is surjective. If n is the sum of the orders of the poles of f in F and $z_0 \in \mathbf{C}$, then $f(z) = z_0$ has n solutions (counting multiplicities).
6. If f has only one pole in F , then this is not a simple pole.

Proof. 1. Let us assume f has infinitely many zeros or poles in any compact set, e.g., $\overline{F}$. Due to compactness of $\overline{F}$, this would result in a limit point of either (say v_0). Then f would be either identically zero (by Identity Theorem) or non holomorphic in a neighbourhood of v_0 . Hence f is non meromorphic, a contradiction. Hence f has only finitely many poles or zeroes.

If f has no poles, $\implies f$ is bounded in $\overline{F} \implies f$ being doubly periodic is bounded in all of $\mathbf{C} \implies f$ is constant by Liouville's Theorem.

2. By Cauchy's Theorem

$$\int_{\partial F} f(z) dz = 2\pi i \sum_{\mathbf{w}} \operatorname{Res}_{\mathbf{w}} f$$

where the line integral is taken counter-clockwise. Now,

$$\int_{\partial F} f(z) dz = \int_0^{\mathbf{w}_2} f(z) dz + \int_{\mathbf{w}_1}^{\mathbf{w}_2 + \mathbf{w}_1} f(z) dz + \int_{\mathbf{w}_1 + \mathbf{w}_1}^{\mathbf{w}_1} f(z) dz + \int_{u_1}^0 f(z) dz$$

$$f(z + \mathbf{w}_1) = f(z) \implies \int_{\mathbf{w}_2 + \mathbf{w}_1}^0 f(z) dz = \int_{\mathbf{w}_2}^0 f(z) dz = - \int_0^{\mathbf{w}_2} f(z) dz.$$

Similarly

$$\int_{\mathbf{w}_2}^{\mathbf{w}_2 + \mathbf{w}_1} f(z) dz = - \int_{\mathbf{w}_1}^0 f(z) dz.$$

Hence the four integrals sum up to 0 .

Remark. A small detour around the poles is made if f has any lying on the path of integration. Result remains same as integrals cancel out [See Fig 3.1].

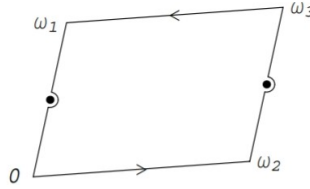


Fig 3.1

3. Let $r = \text{ord}_{\mathbf{w}} f \implies f(z) = (z - w)^r g(z), g(w) \neq 0$ and finite.

$$\begin{aligned} \implies \frac{f'(z)}{f(z)} &= \frac{r}{z - \mathbf{w}} + \frac{g'(z)}{g(z)} \\ \implies \text{Res}_{\mathbf{w}} \left(\frac{f'(z)}{f(z)} \right) &= r \end{aligned}$$

f is doubly periodic $\implies f'$ is doubly periodic. Applying (2.) to f'/f :

$$2\pi i \sum_{w \in F} \text{ord}_w f = 2\pi i \sum_{w \in F} \text{Res}_w \left(\frac{f'}{f} \right) = 0.$$

Hence $\deg(\text{div}(f)) = 0$.

4.

$$2\pi i \sum_{\mathbf{w} \in F} \text{ord}_{\mathbf{w}} f = 2\pi i \sum_{\mathbf{w} \in F} \text{Res}_{\mathbf{w}} \left(\frac{f'(z)}{f(z)} \right) = \int_{\delta F} z \left(\frac{f'(z)}{f(z)} \right) dz$$

But the function zf'/f is not doubly periodic. Hence as in the proof of (2.), The integral can be written as a sum of four integrals. The double periodicity of f and f' gives:

$$\begin{aligned} \int_{\mathbf{w}_1+\mathbf{w}_2}^{\mathbf{w}_1} z \left(\frac{f'(z)}{f(z)} \right) dz &= \int_{\mathbf{w}_2}^0 (z + \mathbf{w}_1) \left(\frac{f'(z)}{f(z)} \right) dz \\ &= - \int_0^{\mathbf{w}_2} z \left(\frac{f'(z)}{f(z)} \right) dz - \mathbf{w}_2 \int_0^{\mathbf{w}_2} \left(\frac{f'(z)}{f(z)} \right) dz. \end{aligned}$$

Now we get $\frac{1}{2\pi i} \int_0^{\mathbf{w}_2} \left(\frac{f'(z)}{f(z)} \right) dz$. This is the winding number around 0 of the path $z = f(t\alpha_2), 0 \leq \alpha \leq 1$. As the path is piecewise smooth, it is an integer. The path is closed as $f(0) = f(\mathbf{w}_2)$.

$$\implies \int_0^{\mathbf{w}_2} z \left(\frac{f'(z)}{f(z)} \right) dz + \int_{\mathbf{w}_1+\mathbf{w}_2}^{\mathbf{w}_1} z \left(\frac{f'(z)}{f(z)} \right) dz = -\mathbf{w}_1 \int_0^{\mathbf{w}_2} \left(\frac{f'(z)}{f(z)} \right) dz \in 2\pi i \mathbf{Z}L$$

Doing similarly for other two integrals, we get $2\pi i \sum_{\mathbf{w} \in F} \mathbf{w} \cdot \text{ord}_{\mathbf{w}} f \in 2\pi i \mathbf{Z}L$.

5. let $z_0 \in C \implies h(z) = f(z) - z_0$ is a doubly periodic function having same poles as h . By (3.), the number of zeros of $h(z)$ in E = number of poles (counting multiplicities) of h , which equals n . This proves (5.).
6. Let f has only one simple pole. Denote it $\mathbf{w}$. $\implies \text{Res}_{\mathbf{w}} f \neq 0. \implies$ the sum in (2.) is nonzero, a contradiction. Hence the pole cannot be simple or there must be other poles present.

□

We now study one more extremely useful example of an elliptic function: the Weierstrass ' $\wp$ ' function.

3.3 Weierstrass $\wp$ -Function

Weierstrass $\wp$ Function. Given a lattice L , define the Weierstrass $\wp$ -function by

$$\wp(z) = \frac{1}{z^2} + \sum_{\substack{\mathbf{w} \in L \\ \mathbf{w} \neq 0}} \frac{1}{(z - \mathbf{w})^2} - \frac{1}{\mathbf{w}^2}.$$

Clearly elements of L are double poles of $\wp$.

Theorem 3.3 $\wp$ function satisfies the following properties:

1. The series-sum of $\wp(z)$ converges absolutely and uniformly, on compact sets not having elements of L .
2. $\wp(-z) = \wp(z)$ for all $z \in \mathbf{C}$.
3. $\wp(z)$ has a double pole at each $\mathbf{w} \in L$ and is also meromorphic in $\mathbf{C}$.
4. $\wp(z + \mathbf{w}) = \wp(z)$ for all $\mathbf{w} \in L$.

Lemma 3.4 If $k > 2$ then $\sum_{\substack{\mathbf{w} \in L \\ \mathbf{w} \neq 0}} \frac{1}{|\mathbf{w}|^k}$ Converges.

Proof can be found in [3].

Proof. of Theorem 3.3

1. Consider as compact set D . Let $M = \text{Max}\{|z| \mid z \in D\}$. If $z \in D$ and $|\mathbf{w}| \geq 2M$,
 $\implies |z - \mathbf{w}| \geq |\mathbf{w}|/2$ and $|2\mathbf{w} - z| \leq 5|\mathbf{w}|/2$, hence

$$\left| \frac{1}{(z - \mathbf{w})^2} - \frac{1}{\mathbf{w}^2} \right| = \left| \frac{z(2\mathbf{w} - z)1}{(z - \mathbf{w})^2 \mathbf{w}^2} \right| \leq \frac{M \left(\frac{5|\mathbf{w}|}{2} \right)}{\frac{|\mathbf{w}^2|}{4}} = \frac{10M}{|\mathbf{w}|^3}.$$

As only finitely many terms have been omitted and by Lemma 3.4, the sum of the terms in above equation with $|\mathbf{w}| \geq 2M$ converges uniformly and also absolutely for $z \in D$.

2. $\mathbf{w} \in L \Leftrightarrow -\mathbf{w} \in L \implies$ in the sum of $\wp(-z)$, we can also sum over $-\mathbf{w} \in L$. The terms of this sum remain same as $\wp(z)$ due to squared terms in the series sum. Therefore $\wp(-z) = \wp(z)$.
3. $\wp(z)$ is clearly holomorphic for $z \notin L$.
 If $z \in L$, then the sum of the terms for $\mathbf{w} \neq z$ is holomorphic near z , and the term $\frac{1}{(z - \mathbf{w})^2}$ make $\wp$ to have a double pole at z . Hence $\wp$ is meromorphic.

4. As the series converges uniformly, we can differentiate $\wp(z)$ term by term to get:

$$\wp'(z) = -2 \sum_{\mathbf{w} \in L} \frac{1}{(z - \mathbf{w})^3}.$$

The sum converges by Lemma 2.1 when $z \notin L$ and $\mathbf{w} = 0$ is included in the sum. Changing z to $z + \mathbf{w}$:

$$\Rightarrow \wp'(z + \mathbf{w}) = \wp'(z) \forall z \notin L$$

$$\implies \wp(z + \mathbf{w}) = \wp(z) + c.$$

Using $\wp(\mathbf{w}_1/2) = 0$, we get $c = 0$.

$$\implies \wp(z + \mathbf{w}) = \wp(z).$$

□

NOTE. From the last part of above Theorem, we get $\wp'$ is also doubly periodic. We see one more related function below.

Proposition 3.5 Let

$$\sigma(z) = \sigma(z; L) = z \prod_{\substack{\mathbf{w} \in \bar{L} \\ \mathbf{w} \neq 0}} \left(1 - \frac{z}{\mathbf{w}}\right) e^{\frac{z}{\mathbf{w}} + \frac{1}{2} \left(\frac{z}{\mathbf{w}}\right)^2}.$$

Then

1. $\sigma(z)$ is holomorphic $\forall z \in \mathbb{C}$.
2. $\sigma(z)$ has simple zeros at each $\mathbf{w} \in L$ and has no other zeros.
3. $\frac{d^2}{dz^2} \log \sigma(z) = -\wp(z)$.
4. given $\mathbf{w} \in L$, $\exists \alpha = \alpha_{\mathbf{w}}$ and $\beta = \beta_{\mathbf{w}}$ satisfying

$$\sigma(z + \mathbf{w}) = e^{\alpha z + \beta} \sigma(z) \forall z \in \mathbb{C}.$$

Proof. 1. Using formula for power series expansion of e^z we get:

$$(1 - u)e^{(u + \frac{1}{2}u^2)} = 1 + c_3u^3 + c_4u^4 + \dots$$

Near $u = 0 \exists C$ (constant), $\left| (1 - u)e^{(u + \frac{1}{2}u^2)} - 1 \right| \leq C|u|^3$. When $u = z/\mathbf{w}$, take sufficiently large $|\mathbf{w}|$ and z . Then this inequality holds.

If the sum $\sum |a_n|$ converges, $\implies \prod (1 + a_n)$ converges [3]. Also $(1 + a_n) \neq 0 \forall n \implies$ the product is nonzero.

$|z/\mathbf{w}|^3$ converges by Lemma 3.4 for $k = 3$. Hence the product defining $\sigma(z)$ converges uniformly on compact sets. $\implies \sigma(z)$ is holomorphic, as limit of uniformly converging holomorphic functions is holomorphic.

2. Obvious from function definition.

3. Logarithmic differentiation of $\sigma(z)$ gives:

$$\frac{d}{dz} \log \sigma(z) = \frac{1}{z} + \sum_{\substack{\mathbf{w} \in L \\ \mathbf{w} \neq 0}} \frac{1}{z - \mathbf{w}} + \frac{z}{\mathbf{w}^2} + \frac{1}{\mathbf{w}^2}.$$

Differentiating above equation gives the required sum for $-\wp(z)$.

4. Let $\mathbf{w} \in L$. Since

$$\frac{d}{dz^2} \log \frac{\sigma(z + \mathbf{w})}{\sigma(z)} = 0.$$

There are constants a, b such that

$$\log \frac{\sigma(z + \mathbf{w})}{\sigma(z)} = az + b.$$

Taking exponent (of e) both sides gets the result.

□

Theorem 3.6 Let $D = \sum m_i [w_i]$ be a divisor. Then D is the divisor of a function $\iff \deg(D) = 0$ and $\sum m_i w_i \in L$.

Proof. ($\implies$) By parts (3.) and (4.) of Theorem 3.2, if D is the divisor of a function then $\deg(D) = 0$ and $\sum m_i w_i \in L$.

($\impliedby$) Let $\deg(D) = 0$. Also let $\sum n_i w_i = s \in L$.

$$\implies \sum n_i = 0 \text{ and } \sum n_i w_i = s. \quad (3.3)$$

Let $f(z) = \frac{\sigma(z)}{\sigma(z-s)} \prod_i \sigma(z - w_i)^{n_i}$.

If $\mathbf{w} \in L$, then by [3.3]

$$\frac{f(z + \mathbf{w})}{f(z)} = e^{a_{\mathbf{w}}z + \beta\mathbf{w} - a_{\mathbf{w}}(z-\ell) - \beta\mathbf{w}} e^{\sum n_i (a_{\mathbf{w}}(z-w_i) + \beta\mathbf{w})} = 1,$$

$\implies$ double periodicity of $f(z)$. Clearly, D is divisor of the function f .

□

This Theorem shall aid us in proving group isomorphism between an Elliptic Curve and a Torus, as seen in the next chapter.

Chapter 4

Tori And Elliptic Curves

4.1 Torus to Elliptic Curve

In this section we see that Elliptic Curves are isomorphic to the Complex Torus, using the $\wp$ function. First note some results.

Eisenstien Series. For integers $k \geq 3$, the series

$$G_k = G_k(L) = \sum_{\substack{\mathbf{w} \neq 0 \\ \mathbf{w} \in L}} \mathbf{w}^{-k}$$

is called the Eisenstein series for the lattice L .

By Lemma 3.4, the series always converges. $\mathbf{w}$ and $-\mathbf{w}$ cancel if k is odd, so $G_k = 0$.

Proposition 4.1

$$\wp(z) = \frac{1}{z^2} + \sum_{j=1}^{\infty} (2j+1)G_{2j+2}z^{2j},$$

for $0 < |z| < \text{Min}_{0 \neq \mathbf{w} \in L}(|\mathbf{w}|)$.

Proof.

$$\begin{aligned}
\frac{1}{(z - \mathbf{w})^2} - \frac{1}{\mathbf{w}^2} &= \mathbf{w}^{-2} \left(\frac{1}{\left(1 - \left(\frac{z}{\mathbf{w}}\right)\right)^2} - 1 \right) \\
&= \mathbf{w}^{-2} \left(\sum_{n=1}^{\infty} (n+1) \left(\frac{z^n}{\mathbf{w}^n} \right) \right) \\
\Rightarrow \wp(z) &= \frac{1}{z^2} + \left(\sum_{\mathbf{w} \neq 0} \sum_{n=1}^{\infty} (n+1) \left(\frac{z^n}{\mathbf{w}^{n+2}} \right) \right)
\end{aligned}$$

Adding the terms first over $\mathbf{w}$ followed by the sum over n gives the result. $\square$

Theorem 4.2 $\wp'(z)^2 = 4\wp(z)^3 - 60G_4\wp(z) - 140G_6$.

Proof. From Proposition 4.1,

$$\wp(z) = z^{-2} + 3G_4z^2 + 5G_6z^4 + \dots \quad (4.1)$$

$$\wp'(z) = -2z^{-3} + 6G_4z + 20G_6z^3 + \dots \quad (4.2)$$

$$\begin{aligned}
\Rightarrow \wp(z)^3 &= z^{-6} + 9G_4z^{-2} + 15G_6 + \dots \\
\wp'(z)^2 &= 4z^{-6} - 24G_4z^{-2} - 80G_6 + \dots
\end{aligned}$$

$$\Rightarrow f(z) = \wp'(z)^2 - 4\wp(z)^3 + 60G_4\wp(z) + 140G_6 = c_1z + c_2z^2 + \dots$$

$\Rightarrow f(z)$ has no pole at zero. The above series also has no constant term. Negative powers of z are also not present. $\Rightarrow$ The only possible poles of $f(z)$ are at the poles of $\wp(z)$ and $\wp'(z)$. These are the elements of L .

Since $f(z)$ is doubly periodic and has no pole at 0 $\Rightarrow f(z)$ has no poles. Hence by Theorem 3.3, $f(z)$ is constant.

Now $f(0) = 0 \Rightarrow f(z)$ is identically 0. Hence we get the result. $\square$

Remark. We denote $g_2 = 60G_4$, $g_3 = 140G_6$. By the above Theorem

$$\wp'(z)^2 = 4\wp(z)^3 - g_2\wp(z) - g_3. \quad (4.3)$$

Hence, the points $(\wp(z), \wp'(z))$ lie on the curve $y^2 = 4x^3 - g_2x - g_3$. The discriminant of the cubic is hence given by $16(g_2^3 - 27g_3^2)$.

Proposition 4.3 $\Delta = g_2^3 - 27g_3^2 \neq 0$.

Proof. By double periodicity of $\wp'(z)$, $\wp'(\mathbf{w}_i/2) = \wp'(-\mathbf{w}_i/2)$.

As $\wp'(-z) = -\wp'(z)$,

$$\implies \wp'(\mathbf{w}_i/2) = 0, \quad i = 1, 2, 3. \quad (4.4)$$

By [4.4] each $\wp(\mathbf{w}_i/2)$ is a root of $4x^3 - g_2x - g_3$. We need to show that the discriminant is nonzero $\implies$ distinct roots of the cubic. Let $h_i(z) = \wp(z) - \wp(\mathbf{w}_i/2)$.

$\implies h_i(\mathbf{w}_i/2) = 0 = h'_i(\mathbf{w}_i/2)$. So $\mathbf{w}_i/2$ is zero of h_i of order at least two.

The double pole of $\wp$ at $z = 0$ is the only pole of $h_i(z)$ in F .

The only zero of $h_i(z)$ is Theorem 3.2 (5.) $\implies \mathbf{w}_i/2$. Particularly $h_i(\mathbf{w}_j/2) \neq 0$, when $j \neq i$. Hence the values $\wp(\mathbf{w}_i/2)$ are distinct and hence discriminant is non-zero. $\square$

Remark. By above proposition, $E : y^2 = 4x^3 - g_2x - g_3$ is the equation of an Elliptic Curve. Hence we get a map from $z \in \mathbf{C}$ to $E(\mathbf{C})$. This is indeed an isomorphism between an Elliptic Curve and a Torus, as proven in the next Theorem.

Theorem 4.4 Let L be a lattice and let E be the Elliptic Curve $y^2 = 4x^3 - g_2x - g_3$. The the mapping

$$\Psi : \begin{cases} \mathbf{C}/L \longrightarrow E(\mathbf{C}) \\ z \longmapsto (\wp(z), \wp'(z)) \\ 0 \longmapsto \infty \end{cases}$$

is an isomorphism between the two groups.

Proof. We prove in 3 steps to show this is an isomorphism:

1. ψ is Surjective

Let $(x, y) \in E(\mathbf{C})$. $\wp(z) - x$ has zeroes as it has a double pole by Theorem 3.2. $\implies \exists z \in \mathbf{C}$ such that $\wp(z) = x$.

By Remark after Theorem 4.2, $\wp'(z)^2 = y^2 \implies \wp'(z) = \pm y$.

If $\wp'(z) = y$, we have found a pre-image. If $\wp'(z) = -y \implies \wp'(-z) = y$ and $\wp(-z) = x$, so $-z \mapsto (x, y)$.

Hence Surjectivity is proven.

2. ψ is Injective

Let $\wp(z_1) = \wp(z_2)$. Also assume $\wp'(z_1) = \wp'(z_2)$, and $z_1 \not\equiv z_2 \pmod{L}$. Following cases arise:

Case-(i) z_1 is a pole: If z_1 is a pole of $\wp$, then $z_1 \in L$ as the only poles of $\wp$ are in L . Hence $z_1 \equiv z_2 \pmod{L}$.

Case-(ii) z_1 is not a pole of $\wp$: Hence $z_1 \notin L$. The only pole of the function (in F)

$$h(z) = \wp(z) - \wp(z_1)$$

is the only double pole at $z = 0$. Let $z_1 = \mathbf{w}_i/2$ for some i . z_1 is the only root of $h(z)$ by [4.4]: which tells that $\wp'(\mathbf{w}_i/2) = 0$. By Theorem 3.2, h has exactly two zeros. Hence $z_2 = z_1$.

If $z_1 \neq \mathbf{w}_i/2 \implies h(-z_1) = h(z_1) = 0$.

$z_1 \not\equiv -z_1 \pmod{L}$, therefore two zeros of h are z_1 and $-z_1 \pmod{L}$. $\implies z_2 \equiv -z_1 \pmod{L}$.

But

$$y = \wp'(z_2) = \wp'(-z_1) = -\wp'(z_1) = -y.$$

$\implies \wp'(z_1) = y = 0$. But $\wp'(z)$ has only one triple pole at 0, hence has exactly three zeros in F by Theorem 3.2. These zeros occur at each $\mathbf{w}_i/2$, by [4.4]. This contradicts our assumption that $z \neq \mathbf{w}_i/2$. $\implies z_1 \equiv z_2 \pmod{L}$.

Hence Injectivity of Ψ is proven.

3. ψ is a Homomorphism

Let $z_1, z_2 \in \mathbf{C}$ and let

$$\Psi(z_i) = P_i = (x_i, y_i).$$

Let P_1, P_2 be finite points of E . Hence $P_1 \neq \pm P_2$, that $2P_1 + P_2 \neq \infty$, and that $P_1 + 2P_2 \neq \infty$.

Let $y = ax + b$ be the line through P_1, P_2 . Let $P_3 = (x_3, y_3)$ denote the point of intersection of this line with E . Let $(x_3, y_3) = P_3 = \Psi(z_3)$, $z_3 \in \mathbf{C}$. By GROUP LAW:

$$x_3 = \frac{1}{4} \left(\frac{y_2 - y_1}{x_2 - x_1} \right)^2 - x_1 - x_2 = \frac{1}{4} \left(\frac{\wp'(z_2) - \wp'(z_1)}{\wp(z_2) - \wp(z_1)} \right)^2 - \wp(z_1) - \wp(z_2). \quad (4.5)$$

Consider the function

$$k(z) = \wp'(z) - a\wp(z) - b.$$

It has zeroes at $z = z_1, z_2, z_3$ (From $y = \wp'(z)$, and by our assumption $y = ax + b$). Since 0 is the only pole (triple pole precisely), $k(z)$ has three zeros in F . Hence Theorem 3.6 gives

$$\operatorname{div}(k) = [z_1] + [z_2] + [z_3] - 3[0].$$

Now $z_1 + z_2 + z_3 \in L$.

$$\implies \wp(z_1 + z_2) = \wp(-z_3) = \wp(z_3) = x_3.$$

Hence [4.5] gives

$$\wp(z_1 + z_2) = \frac{1}{4} \left(\frac{\wp'(z_2) - \wp'(z_1)}{\wp(z_2) - \wp(z_1)} \right)^2 - \wp(z_1) - \wp(z_2) \quad (4.6)$$

We now consider the y -coordinate. We find $\wp'(z_1 + z_2)$. Differentiate [4.6] with respect to z_2 . This expresses $\wp'(z_1 + z_2)$ in terms of x_1, x_2, y_1, y_2 , and $\wp''(z_2)$. We need to express $\wp''$ in terms of $\wp$ and $\wp'$. Differentiating [4.3] gives

$$\wp''\wp' = (12\wp^2 - g_2)\wp'. \quad (4.7)$$

Divide by $\wp'(z)$ (if $\wp'(z) \neq 0$; the other cases are due to continuity of $\wp'$ as it is meromorphic) to get

$$\wp''(z_2) = 12\wp(z_2)^2 - g_2.$$

Substituting this in expansion for $\wp'(z_1 + z_2)$ gives $\wp'(z_1 + z_2)$ in terms of $\wp(z_1), \wp'(z_1), \wp(z_2), \wp'(z_2)$.

Doing some algebraic manipulation, we get to know this is $= -y_3$ obtained from the GROUP LAW for $(x_1, y_1) + (x_2, y_2) = (x_3, -y_3)$. Hence,

$$(\wp(z_1), \wp'(z_1)) + (\wp(z_2), \wp'(z_2)) = (\wp(z_1 + z_2), \wp'(z_1 + z_2)).$$

$$\implies \Psi(z_1) + \Psi(z_2) = \Psi(z_1 + z_2). \quad (4.8)$$

We now check [4.8] in the cases where [4.6] is not defined.

The cases where $\wp(z_i) = \infty$ and where $z_1 \equiv -z_2 \pmod{L}$ can be easily checked.

The remaining case is for $z_1 = z_2$. Let $z_2 \rightarrow z_1$ in [4.6]. Use l'Hôpital's rule and [4.7] to obtain

$$\begin{aligned}\wp(2z_1) &= \frac{1}{4} \left(\frac{\wp''(z_1)}{\wp'(z_1)} \right)^2 - 2\wp(z_1) \\ &= \frac{1}{4} \left(\frac{6\wp(z_1)^2 - \frac{1}{2}g_2}{\wp'(z_1)} \right)^2 - 2\wp(z_1) \\ &= \frac{1}{4} \left(\frac{6x_1^2 - \frac{1}{2}g_2}{y_1} \right)^2 - 2x_1.\end{aligned}$$

This is the formula for x_3 as per the GROUP LAW. The formula for the y_3 comes by differentiating w.r.t. z_1 gives .

$$\implies \Psi(z_1) + \Psi(z_1) = \Psi(2z_1).$$

$\implies \Psi$ is a Group Homomorphism. □

4.2 Elliptic Curve To Torus

In this section, we shall prove that for every Elliptic Curve there corresponds a lattice $L = w_1\mathbf{Z} + w_2\mathbf{Z}$ or it comes from the Torus $\mathbf{C}/L$. First note some definitions.

Homothetic Lattice. A lattice $L_\beta = \beta\mathbf{Z} + \mathbf{Z}$ is homothetic to L if there exists a nonzero complex number μ satisfying $L = \mu L_\beta$.

Example. Let $L = \mathbf{Z}\mathbf{w}_1 + \mathbf{Z}\mathbf{w}_2$ and $\beta = \mathbf{w}_1/\mathbf{w}_2$. $\mu = \mathbf{w}_2$ here.

Remark. $\mathbf{w}_1$ and $\mathbf{w}_2$ are linearly independent over $\mathbf{R} \implies \beta \notin \mathbf{R}$. We interchange $\mathbf{w}_1$ and $\mathbf{w}_2$ in β (if required) to keep $\Im(\beta) > 0$.

Eisenstien Series For A Lattice. For an integer $k \geq 3$, define

$$G_k(\beta) = G_k(L_\beta) = \sum_{(m,n) \neq (0,0)} \frac{1}{(m\beta + n)^k}.$$

We have

$$G_k(\beta) = \mathbf{w}_2^k G_k(L)$$

where $L = \mathbf{Z}\mathbf{w}_1 + \mathbf{Z}\mathbf{w}_2$.

We state a corollary to be used. Proof can be found in [3].

Corollary 4.5 $\forall z \in \mathbf{C} \exists$ exactly one $\beta \in F$ such that $j(\beta) = z$.

Theorem 4.6 Let $y^2 = 4x^3 - Ax - B$ define an Elliptic Curve E over $\mathbf{C}$. Then there is a lattice L satisfying

$$g_2(L) = A \quad \text{and} \quad g_3(L) = B.$$

$\mathbf{C}/L \simeq E(\mathbf{C})$ is a group isomorphism.

Proof. Let

$$j = 1728 \frac{A^3}{A^3 - 27B^2}.$$

By Corollary 4.5, $j(\beta) = j(L) = j$ and there exists a lattice $L_\beta = \mathbf{Z}\beta + \mathbf{Z}$. We have the following cases:

Case-(i) $g_2(L) \neq 0$:

Hence $j = j(L) \neq 0 \implies A \neq 0$. Choose a $\beta \in \mathbf{C}^\times$ satisfying

$$g_2(\beta L) = \beta^{-4} g_2(L) = A.$$

$j = j(L)$ gives

$$g_3(\beta L)^2 = B^2,$$

so $g_3(\beta L) = \pm B$.

If $g_3(\beta L) = B$, proof done.

If $g_3(\beta L) = -B$, then

$$g_3(i\beta L) = i^{-6} g_3(\beta L) = B \quad \text{and} \quad g_2(i\beta L) = i^4 g_2(\beta L) = A.$$

Hence either βL or $i\beta L$ the required lattice.

Case-(ii) If $g_2(L) = 0$:

$\implies j = j(L) = 0 \implies A = 0$. $A^3 - 27B^2 \neq 0$. Also $g_2(L)^3 - 27g_3(L)^2 \neq 0$ by Proposition 4.3. Hence $B \neq 0$ and $g_3(L) \neq 0$. Choose $\beta \in \mathbf{C}^\times$ (non-zero β) satisfying

$$g_3(\beta L) = \beta^{-6} g_3(L) = B.$$

Then $g_2(\beta L) = \beta^{-4} g_2(L) = 0 = A$, so βL is the required lattice.

$\mathbf{C}/L \longrightarrow E(\mathbf{C})$ is a group isomorphism by Theorem 4.4. $\square$

Remark. The elements $\mathbf{w}_1$ and $\mathbf{w}_2$ of L are called the periods L . In the next chapter we shall see how to compute them with the help of elliptic integrals.

Chapter 5

Elliptic Integrals

5.1 Elliptic Integrals

Given $E(\mathbf{R})$

$$y^2 = 4x^3 - g_2x - g_3 = 4(x - \rho_1)(x - \rho_2)(x - \rho_3), \quad \rho_1 < \rho_2 < \rho_3.$$

and the integral

$$\int_{\rho_3}^{\infty} \frac{dx}{\sqrt{4(x - \rho_1)(x - \rho_2)(x - \rho_3)}},$$

the change of variable $x = \wp(z)$ gives:

$$\implies \int_0^{\mathbf{w}_2/2} dz = \frac{\mathbf{w}_2}{2}$$

as denominator becomes $\sqrt{y^2} = \sqrt{\wp'(z)^2} = \wp'(z)$. See proof of Proposition 4.3 for choosing this upper limit.

$$\implies \mathbf{w}_2 = \int_{\rho_3}^{\infty} \frac{dx}{\sqrt{(x - \rho_1)(x - \rho_2)(x - \rho_3)}}.$$

The variable transformation

$$x = \frac{\left(\rho_3 - \sqrt{(\rho_3 - \rho_1)(\rho_3 - \rho_2)}\right)t + \left(\rho_3 + \sqrt{(\rho_3 - \rho_1)(\rho_3 - \rho_2)}\right)}{t + 1}$$

transforms the above integral to

$$\begin{aligned} w_2 &= \frac{2}{\sqrt{\rho_3 - \rho_1} + \sqrt{\rho_1 - \rho_2}} \int_{-1}^1 \frac{dt}{\sqrt{(1-t^2)(1-k^2t^2)}} \\ &= \frac{4}{\sqrt{\rho_3 - \rho_1} + \sqrt{\rho_3 - \rho_2}} \int_0^1 \frac{dt}{\sqrt{(1-t^2)(1-k^2t^2)}}. \end{aligned}$$

Here

$$k = \frac{\sqrt{\rho_3 - \rho_1} - \sqrt{\rho_3 - \rho_2}}{\sqrt{\rho_3 - \rho_2} + \sqrt{\rho_3 - \rho_2}}.$$

We denote

$$K(k) = \int_0^1 \frac{dt}{\sqrt{(1-t^2)(1-k^2t^2)}}.$$

This is an **Elliptic Integral**. In the next section, we study the sequence called the **arithmetic-geometric mean**. Calculating its limit gives us a very fast method for computing elliptic integrals to a given accuracy.

5.2 Arithmetic-Geometric Mean

Consider two real numbers $\alpha, \beta \in \mathbf{R}^+$. Define α_n and β_n as

$$\begin{aligned} \alpha_0 &= \alpha, & \beta_0 &= \beta \\ \alpha_n &= \frac{1}{2}(\alpha_{n-1} + \beta_{n-1}) \\ \beta_n &= \sqrt{\alpha_{n-1}\beta_{n-1}}. \end{aligned} \tag{5.1}$$

Then α_n is the **arithmetic mean** of α_{n-1} and β_{n-1} , and β_n is their **geometric mean**.

Proposition 5.1 Suppose $\alpha \geq \beta \geq 0$. Then

$$\beta_{n-1} \leq \beta_n \leq \alpha_n \leq \alpha_{n-1}$$

and

$$0 \leq \alpha_n - \beta_n \leq \frac{1}{2}(\alpha_{n-1} - \beta_{n-1}). \tag{5.2}$$

Therefore

$$\lim_{n \rightarrow \infty} \alpha_n = \lim_{n \rightarrow \infty} \beta_n = M(a, b)$$

exists. If $b \geq 1$ then

$$\alpha_{n+m} - \beta_{n-m} \leq 8 \left(\frac{\alpha_n - \beta_n}{8} \right)^{2^{2m}} \quad (5.3)$$

for all $m, n \geq 0$.

The limit $M(\alpha, \beta)$ is called the **arithmetic-geometric mean** of α and β .

Proof. $\alpha_n \geq \beta_n \forall n \in \mathbf{N}$, as $\alpha_n - \beta_n = \frac{1}{2} \left(\sqrt{\alpha_{n-1}} - \sqrt{\beta_{n-1}} \right)^2 \geq 0$.

[5.1] implies that:

$$\alpha_n \leq \frac{1}{2} (\alpha_{n-1} + \alpha_{n-1}) = \alpha_{n-1} \quad \text{and} \quad \beta_n \geq \sqrt{\beta_{n-1}\beta_{n-1}} = \beta_{n-1}.$$

$$\implies \beta_{n-1} \leq \beta_n \leq \alpha_n \leq \alpha_{n-1}.$$

Now,

$$\alpha_n - \beta_n = \frac{1}{2} \left(\sqrt{\alpha_{n-1}} - \sqrt{\beta_{n-1}} \right)^2 \leq \frac{1}{2} \left(\sqrt{\alpha_{n-1}} - \sqrt{\beta_{n-1}} \right) \left(\sqrt{\alpha_{n-1}} + \sqrt{\beta_{n-1}} \right) = \frac{1}{2} (\alpha_{n-1} - \beta_{n-1})$$

Continuing similarly, we get $\alpha_n - \beta_n \leq (1/2)^n (\alpha - \beta)$, so $\alpha_n - \beta_n \rightarrow 0$ as $n \rightarrow \infty$.

α_n is a decreasing sequence bounded below by β_n : an increasing sequence. Hence the two sequences converge to the same limit by limit comparison results. Hence $M(a, b)$ exists.

Now if $\beta_{n-1} \geq 1$, as $\alpha_{n-1} \geq \beta_{n-1} \implies \sqrt{\alpha_{n-1}} + \sqrt{\beta_{n-1}} \geq 2$,

$$\begin{aligned} \implies \frac{\alpha_n - \beta_n}{8} &= \frac{1}{16} \left(\sqrt{\alpha_{n-1}} - \sqrt{\beta_{n-1}} \right)^2 \leq \frac{1}{16} \left(\sqrt{\alpha_{n-1}} - \sqrt{\beta_{n-1}} \right)^2 \frac{\left(\sqrt{\alpha_{n-1}} + \sqrt{\beta_{n-1}} \right)^2}{4} \\ &= \left(\frac{\alpha_{n-1} - \beta_{n-1}}{8} \right)^2. \end{aligned}$$

Inequality [5.3] can be proved from induction. □

Remark. Note that $M(c\alpha, c\beta) = cM(\alpha, \beta)$. By Inequality [5.3], $\alpha_n - \beta_n < 1$ for sufficiently large n and the numbers α_{n+m} and β_{n+m} give best approximations to $M(\alpha, \beta)$. By inequality [5.3], the number of decimal places of accuracy doubles with each iteration [3].

The following two propositions, which help in calculating elliptic integrals use this limit

arithmetic-geometric mean.

Proposition 5.2 Let a, b be positive real numbers. Define

$$I(a, b) = \int_0^{\pi/2} \frac{d\phi}{\sqrt{a^2 \cos^2 \phi + b^2 \sin^2 \phi}}.$$

Then

$$I\left(\frac{a+b}{2}, \sqrt{ab}\right) = I(a, b).$$

Moreover,

$$I(a, b) = \frac{\pi/2}{M(a, b)}.$$

Proof. By the tranformation $u = b \tan \phi$, the integral changes to:

$$\begin{aligned} I(a, b) &= \int_0^\infty \frac{du}{\sqrt{(u^2 + a^2)(u^2 + b^2)}} = \frac{1}{2} \int_{-\infty}^\infty \frac{du}{\sqrt{(u^2 + a^2)(u^2 + b^2)}}. \\ \implies I\left(\frac{a+b}{2}, \sqrt{ab}\right) &= \frac{1}{2} \int_{-\infty}^\infty \frac{du}{\sqrt{\left(u^2 + \left(\frac{a+b}{2}\right)^2\right)(u^2 + ab)}}. \end{aligned}$$

Let

$$u = \frac{1}{2} \left(v - \frac{ab}{v} \right), \quad (0 < v < \infty) \implies v = u + \sqrt{u^2 + ab}.$$

Because

$$\begin{aligned} u^2 + \left(\frac{a+b}{2}\right)^2 &= \frac{1}{4v^2} (v^2 + a^2)(v^2 + b^2) \\ \implies I\left(\frac{a+b}{2}, \sqrt{ab}\right) &= \int_0^\infty \frac{dv}{\sqrt{(v^2 + a^2)(v^2 + b^2)}} = I(a, b). \end{aligned}$$

And by induction:

$$I(a, b) = I(\alpha_1, \beta_1) = I(\alpha_2, \beta_2) = \dots$$

Let

$$\alpha_\infty = \beta_\infty = M(a, b) = \lim_{n \rightarrow \infty} \alpha_n = \lim_{n \rightarrow \infty} \beta_n.$$

Due to uniform convergence of the integrand, we can interchange the limit and integral.

$$\implies I(a, b) = \lim_{n \rightarrow \infty} I(\alpha_n, \beta_n) = I(\alpha_\infty, \beta_\infty) = \int_0^{\pi/2} \frac{d\phi}{\sqrt{\alpha_\infty^2 \cos^2 \phi + \beta_\infty^2 \sin^2 \phi}}$$

$$= \frac{1}{M(a, b)} \int_0^{\pi/2} \frac{d\phi}{\sqrt{\cos^2 \phi + \sin^2 \phi}} = \frac{\pi/2}{M(a, b)}.$$

□

Proposition 5.3 If $0 < k < 1$, then

$$K(k) = I\left(1, \sqrt{1 - k^2}\right) = I(1 + k, 1 - k).$$

Proof.

$$\begin{aligned} K(k) &= \int_0^1 \frac{dt}{\sqrt{(1 - t^2)(1 - k^2 t^2)}} = \int_0^{\pi/2} \frac{d\phi}{\sqrt{1 - k^2 \sin^2 \phi}} \quad (\text{let } t = \sin \phi) \\ &= \int_0^{\pi/2} \frac{d\phi}{\sqrt{\cos^2 \phi + (1 - k^2) \sin^2 \phi}} = I(1, \sqrt{1 - k^2}) = I(1 + k, 1 - k). \end{aligned}$$

The last term of the above equation is by taking $a = 1 + k$ and $b = 1 - k$ in Proposition 5.2. □

5.3 Computing Periods

Finally we compute $\mathbf{w}_1$ and $\mathbf{w}_2$.

Theorem 5.4 Let

$$E : y^2 = 4x^3 - g_2x - g_3 = 4(x - \rho_1)(x - \rho_2)(x - \rho_3)$$

with real numbers $\rho_1 < \rho_2 < \rho_3$. Then $\mathbf{Z}\mathbf{w}_1 + \mathbf{Z}\mathbf{w}_2$ is the required lattice for E , with

$$\begin{aligned} \mathbf{w}_1 &= \frac{\pi i}{M(\sqrt{\rho_3 - \rho_1}, \sqrt{\rho_2 - \rho_1})} \\ \mathbf{w}_2 &= \frac{\pi}{M(\sqrt{\rho_3 - \rho_1}, \sqrt{\rho_3 - \rho_2})}. \end{aligned}$$

Proof. We have, with definition of k as in Section 5.1.

$$\begin{aligned} \mathbf{w}_2 &= \frac{4}{\sqrt{\rho_3 - \rho_1} + \sqrt{\rho_3 - \rho_2}} K(k) \\ &= \frac{4}{\sqrt{\rho_3 - \rho_1} + \sqrt{\rho_3 - \rho_2}} I(1 + k, 1 - k). \end{aligned}$$

The formula for k and $cI(ca, cb) = c \frac{\pi/2}{M(ac, cb)} = I(a, b)$ (from Proposition 5.2) with

$$r = \frac{\sqrt{\rho_3 - \rho_1} + \sqrt{\rho_3 - \rho_2}}{2}$$

gives

$$\begin{aligned} \mathbf{w}_2 &= 2I(\sqrt{\rho_3 - \rho_1}, \sqrt{\rho_3 - \rho_2}) \\ &= \frac{\pi}{M(\sqrt{\rho_3 - \rho_1}, \sqrt{\rho_3 - \rho_2})}. \end{aligned}$$

Similarly

$$\begin{aligned} \mathbf{w}_1 &= \frac{2i}{\sqrt{\rho_3 - \rho_1} + \sqrt{\rho_3 - \rho_2}} K(\sqrt{1 - k^2}) = \frac{2i}{\sqrt{\rho_3 - \rho_1}, \sqrt{\rho_3 - \rho_2}} I(1, k) \\ &= 2iI(\sqrt{\rho_3 - \rho_1} + \sqrt{\rho_3 - \rho_2}, \sqrt{\rho_3 - \rho_1} - \sqrt{\rho_3 - \rho_2}). \end{aligned}$$

If

$$a = \sqrt{\rho_3 - \rho_1} + \sqrt{\rho_3 - \rho_2}, \quad b = \sqrt{\rho_3 - \rho_1} - \sqrt{\rho_3 - \rho_2}$$

then [5.1] gives

$$\alpha_1 = \sqrt{\rho_3 - \rho_1}, \quad \beta_1 = \sqrt{\rho_2 - \rho_1}.$$

hence by Proposition 5.2:

$$\begin{aligned} \mathbf{w}_1 &= 2iI(\sqrt{\rho_3 - \rho_1}, \sqrt{\rho_2 - \rho_1}) \\ &= \frac{\pi i}{M(\sqrt{\rho_3 - \rho_1}, \sqrt{\rho_2 - \rho_1})}. \end{aligned}$$

□

Chapter 6

Application of Elliptic Integral in Finding A Conformal Mapping

6.1 Schwarz Christoffel Transformation

Schwarz–Christoffel Transformation is a conformal mapping of the upper half-plane onto the interior of a simple polygon. We explain its construction in this section.

6.1.1 Construction of One-One Analytic Function

1. Simple Function Involving a Point And an Angle

Let x_1 be a point on the real axis and $f(z)$ be a function such that its derivative is

$$f'(z) = (z - x_1)^k,$$

where $k \in \mathbf{R}$ and $-1 < k < 1$. We Choose $-\pi/2 < \arg(z - x_1) < 3\pi/2$. Hence, there exists a branch cut vertically downward from x_1 .

Nature of the Image of Real Axis Under This Mapping f

Consider any z that lies in the real axis to the left of x_1 , i.e., $z < x_1$. Multiples of 2π are ignored in this derivation.

Observations:

1. As f' exists and is non-zero $\implies f$ is conformal at z by Theorem 1.1.
2. $\arg f'(z) = \arg (z - x_1)^k = k \arg (z - x_1) = k\pi$. $\implies \arg f'(z)$ is constant for all such z .
3. $f'(z) = (z - x_1)^k$ maps the portion of the real line $(-\infty, x_1)$, onto portion of a straight line from ∞ to the point $f(x_1)$ (passing through origin) whose tangents make an angle $k\pi$.
4. Now consider any z to the right of $x_1 : x_1 < z$.

$$\arg f'(z) = \arg (z - x_1)^k = k \cdot 0 = 0.$$

. Therefore point remains on real axis only.

2. Next Level Function Having One More Parameter

Consider

$$f'(z) = A(z - x_1)^k,$$

Here $A \neq 0$ and $k \in \mathbf{R}$ such that $-1 < k < 1$.

$$\implies \arg f'(z) = \arg A + k \arg (z - x_1).$$

Observations

- For the interval $(-\infty, x_1)$, the image is the straight line from ∞ to $f(x_1)$ passing through the origin, whose tangents make an angle $k\pi + \arg A$.
- For the interval (x_1, ∞) , the image is the straight line from $f(x_1)$ to ∞ (not passing through the origin), whose tangents make an angle $\arg A$.
- The angle of the turn at $f(x_1)$ remains unchanged.

3. The General Function

Consider the function

$$f'(z) = A(z - x_1)^{k_1} (z - x_2)^{k_2} \cdots (z - x_n)^{k_n}$$

where $A \neq 0 \in \mathbf{C}$, each k_i satisfies $-1 < k_i < 1$ ($i = 1, 2, \dots, n$) and the x_i (that are real) satisfy $x_1 < x_2 < \cdots < x_n$. Note $\arg z - x_i, i = 1, 2, \dots, n$ lies between $-\pi/2$ and $3\pi/2$. From the above expression, we have

$$\arg f'(z) = \arg A + k_1 \arg (z - x_1) + k_2 \arg (z - x_2) + \cdots + k_n \arg (z - x_n).$$

For each interval $(-\infty, x_1), (x_1, x_2), \dots, (x_{n-1}, x_n)$ the images are straight lines making particular angles measured counterclockwise from the horizontal, given as follows:

^c Interval	Angle of Image
$(-\infty, x_1)$	$\arg A + k_1\pi + k_2\pi + \cdots + k_n\pi$
(x_1, x_2)	$\arg A + k_2\pi + \cdots + k_n\pi$
(x_1, x_2)	$\arg A + k_2\pi + \cdots + k_n\pi$
$\vdots$	$\vdots$
(x_{n-1}, x_n)	$\arg A + k_n\pi$
(x_n, ∞)	$\arg A.$

As z traverses to the right from $-\infty$ through each point $x_1, x_2, \dots, x_n$ and goes to ∞ , the image of $f(z)$ generates a polygonal path whose tangent at the point $f(x_i)$ makes a right turn through the angle $k_i\pi$.

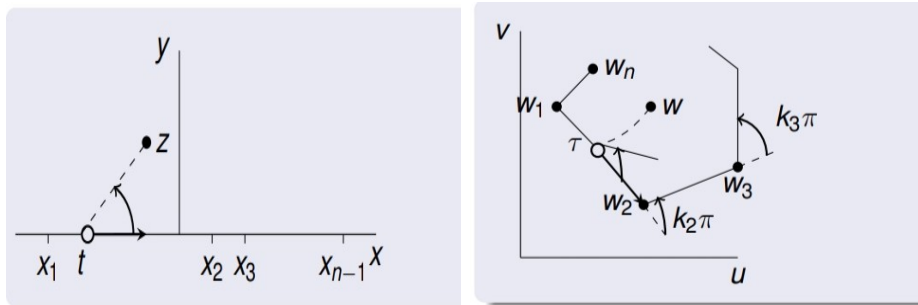


Fig 6.1 $f(x_i) = w_i$

Analytic Behaviour of f

$f'(z) = A(z - x_1)^{k_1} (z - x_2)^{k_2} \cdots (z - x_n)^{k_n}$, hence f is differentiable on the complex plane, except at the branch cuts of the points $x_i, i = 1, 2, \dots, n$.

$\Rightarrow$ The function f is holomorphic in the upper half plane. For any z in the upper half plane define taking γ as the line segment from 0 to z

$$h(z) := \int_{\gamma} f'(\mathbf{w}) d\mathbf{w},$$

$$\Rightarrow f(z) = A \int_0^z (\mathbf{w} - x_1)^{k_1} (\mathbf{w} - x_2)^{k_2} \cdots (\mathbf{w} - x_n)^{k_n} d\mathbf{w} + B.$$

Functions satisfying $f(z)$ are called **Schwarz-Christoffel Transformations**. These transformations map the real axis onto a polygonal path (boundary of the polygon).

6.1.2 Construction Of Conformal Mapping

Let P be the polygon having vertices at the consecutive points $\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_n$ taken in the counterclockwise order. Therefore P has positive orientation. While traversing the polygon, we make a right turn at each vertex $\mathbf{w}_i$ through an angle ω_i and $-\pi < \omega_i < \pi$. A negative value of ω_i indicates a left turn at that position. The total rotation for a counterclockwise tour (travel) must be 2π radians to the left, because

$$\omega_1 + \omega_2 + \cdots + \omega_n = -2\pi.$$

Mapping Real Axis to the Polygon P

We map the real axis under the Schwarz-Christoffel transformation $w = g(z)$.

Choose $n-1$ points, $x_1, x_2, \dots, x_{n-1}$. These $n-1$ points are pre-images of $\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_{n-1}$, respectively. Let both $x = -\infty$ and $x = \infty$ be pre-images of $\mathbf{w}_n$. The function

$$g(z) = \int_0^z (\mathbf{w} - x_1)^{\omega_1/\pi} (\mathbf{w} - x_2)^{\omega_2/\pi} \cdots (\mathbf{w} - x_n)^{\omega_n/\pi} d\mathbf{w},$$

maps the real axis onto some polygon P' , which has the same behaviour of angles at the vertices of the polygon as P , hence P' can be converted into P by some fractional linear transformation¹ as they are topologically equivalent.

¹refer Chapter-7 Of [2]

Hence we have:

Theorem 6.1 Let P be a positively oriented polygon having consecutive corners labeled as $\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_n$ with corresponding angles $\omega_1, \omega_2, \dots, \omega_n$, respectively. Let $x_1, x_2, \dots, x_n$ are points on the real axis (boundary of upper half plane). Then there exists a function of the form given by

$$f(z) = A \int_0^z (\mathbf{w} - x_1)^{\omega_1/\pi} (\mathbf{w} - x_2)^{\omega_2/\pi} \dots (\mathbf{w} - x_n)^{\omega_n/\pi} d\mathbf{w} + B \quad (6.1)$$

(here $A, B \in \mathbf{C}$) that conformally maps the upper half plane into the interior of the polygon P and such a mapping is one-to-one. In addition to that this function satisfies $f(x_1) = \mathbf{w}_1, f(x_2) = \mathbf{w}_2, \dots, f(x_{n-1}) = \mathbf{w}_{n-1}, f(\infty) = \mathbf{w}_n$.

NOTE

1. To construct such a mapping we need value of the function at three points. Since, $f(\infty) = \mathbf{w}_n$, it is enough to know the behaviour at two different points only to determine all the other $n - 3$ points.
2. Even though, the solution is given in $f(z) = g(z) + B$, it is not easy to solve such integral. We shall be seeing in the following section how elliptic integrals can be useful in computing the same.

6.2 Conformal Mapping of A Rectangle to Upper half Plane

Suppose that a rectangle with vertices at $u = a, a + ib, -a + ib, -a$ is given where a and b are positive real numbers.

If we denote by $c_i (i = 1, 2, 3, 4)$ the points of the real axis that are the images of the vertices of the rectangle, then by taking $\phi = -\pi/2$ in [6.1] we get:

$$C'u + C'' = \int_0^2 \frac{dx}{\sqrt{(x - c_1)(x - c_2)(x - c_3)(x - c_4)}}.$$

We require that the points $z = -1, 0$, and 1 correspond to the points $u = -a, 0$, and a . Using this, we determine three of the constants constants as $C''' = 0, c_3 = -1$, and $c_4 = 1$.

$$\implies C'u = \int_0^z \frac{dx}{\sqrt{(x^2 - 1)(x - c_1)(x - c_2)}}. \quad (6.2)$$

Replacing u by $-u$ and z by $-z$ in [6.2]:

$$\begin{aligned} -C'u &= \int_0^{-z} \frac{dx}{\sqrt{(x^2 - 1)(x - c_1)(x - c_2)}} \\ \implies C'u &= \int_0^z \frac{dx}{\sqrt{(x^2 - 1)(x + c_1)(x + c_2)}}. \end{aligned} \quad (6.3)$$

Because of uniqueness of the mapping function under the assumed three boundary points, the function [6.2] must be identical with [6.3].

$$\implies (x - c_1)(x - c_2) = (x + c_1)(x + c_2), \implies c_2 = -c_1.$$

And

$$(x - c_1)(x - c_2) = x^2 - c_1^2,$$

Replace c_1 by $1/k$, where $0 < k < 1$. The mapping function is represented as:

$$u = \frac{1}{C} \int_0^z \frac{dx}{\sqrt{(1 - x^2)(1 - k^2 x^2)}}. \quad (6.4)$$

We now have only two parameters C and k . To determine them we can solve:

$$a = \frac{1}{C} \int_0^1 \frac{dx}{\sqrt{(1 - x^2)(1 - k^2 x^2)}}, \quad (6.5)$$

$$bi = \frac{i}{C} \int_1^{1/k} \frac{dx}{\sqrt{(x^2 - 1)(1 - k^2 x^2)}}. \quad (6.6)$$

Dividing the second by the first, we get for k :

$$\frac{b}{a} = \int_1^{1/k} \frac{dx}{\sqrt{(x^2 - 1)(1 - k^2 x^2)}} / \int_0^1 \frac{dx}{\sqrt{(1 - x^2)(1 - k^2 x^2)}}. \quad (6.7)$$

Once k is determined, C can be found.

We transform the integral

$$\int_1^{1/k} \frac{dx}{\sqrt{(x^2 - 1)(1 - k^2 x^2)}}$$

using

$$k^2 x^2 + k'^2 y^2 = 1$$

assuming that $0 \leq y \leq 1$ when $1/k \geq x \geq 1$. Thus, [6.7] becomes

$$\frac{b}{a} = \int_0^1 \frac{dx}{\sqrt{(1 - x^2)(1 - k'^2 x^2)}} / \int_0^1 \frac{dx}{\sqrt{(1 - x^2)(1 - k^2 x^2)}} = K'/K. \quad (6.8)$$

On the right-hand side we have the ratio of two elliptic integrals K' and K , which can be calculated using formulae of Chapter-5 if k is known.

Now the right-hand side varies monotonically from ∞ to 0 as k increases from 0 to 1. Hence for a given (arbitrary) value of the ratio $b/a \exists k \in (0, 1)$ such that [6.8] holds.

Hence by [6.4], [6.5], [6.6], [6.7], [6.8] and above result the mapping function is:

$$u(z) = \frac{a}{K} \int_0^z \frac{dx}{\sqrt{(1 - x^2)(1 - k^2 x^2)}}.$$

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